10/29/2026

Last time:

Thm (Euler) The series

$$\sum_p \frac{1}{p}$$

diverges. In particular, there are infinitely many primes.

Our proof gives more:

$$\sum_{p \le N} \frac{1}{p} > \ln(\ln N) - C, \quad N \ge 3,$$

where

$$C = \sum_p \frac{1}{p^2} < \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

$$\frac{\pi^2}{6}$$

Define the *prime counting function*.

$$\pi(N) = \text{number of primes} \leq N.$$

In fact, we may extend it to

$$\pi(x) = \text{number of primes} \leq x, \quad x > 0.$$

$$= \pi(\lfloor x \rfloor).$$

We already know

$$\lim_{N \to \infty} \pi(N) = \infty.$$

Q: How fast does $\pi(N)$ grow?

$$\pi(N) \leq 1 + \frac{N-1}{2} = \frac{N+1}{2}, \quad N \geq 2.$$

↑ 2 is prime

↑ number of odd numbers in $(1, N]$

The prime counting function

$\pi(x) = \#\{p \le x : p \text{ is prime}\}$

$\ln x$ denotes the natural logarithm

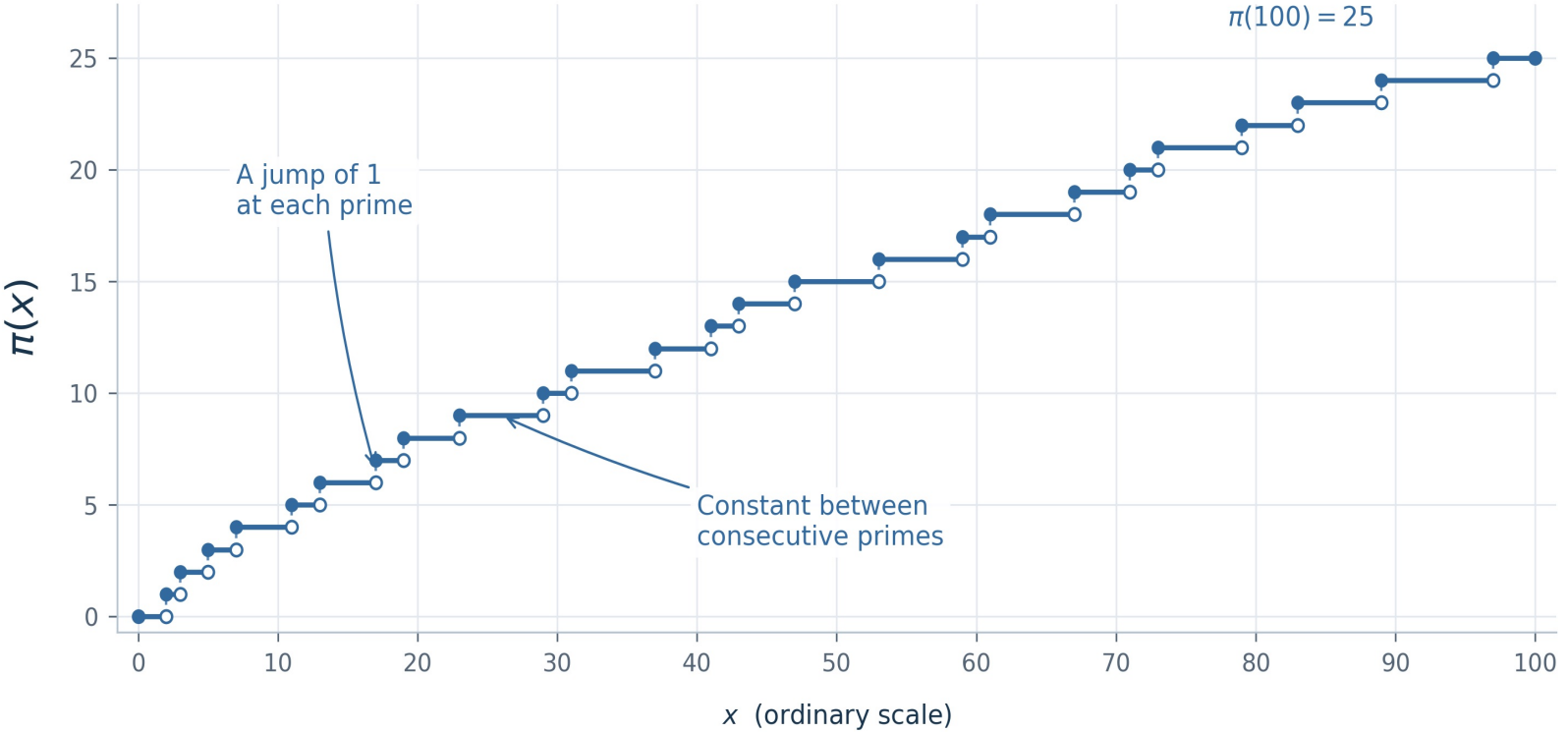
x	$\pi(x)$	$\frac{\pi(x)}{x}$	$\frac{\pi(x)}{x/\ln x}$
10^1	4	0.400000	0.921034
10^2	25	0.250000	1.151293
10^3	168	0.168000	1.160503
10^4	1,229	0.122900	1.131951
10^5	9,592	0.095920	1.104320
10^6	78,498	0.078498	1.084490
10^7	664,579	0.066458	1.071175
10^8	5,761,455	0.057615	1.061299
10^9	50,847,534	0.050848	1.053727
10^{10}	455,052,511	0.045505	1.047797
10^{11}	4,118,054,813	0.041181	1.043039
10^{12}	37,607,912,018	0.037608	1.039145
10^{13}	346,065,536,839	0.034607	1.035899
10^{14}	3,204,941,750,802	0.032049	1.033151
10^{15}	29,844,570,422,669	0.029845	1.030795
10^{16}	279,238,341,033,925	0.027924	1.028752
10^{17}	2,623,557,157,654,233	0.026236	1.026964
10^{18}	24,739,954,287,740,860	0.024740	1.025385
10^{19}	234,057,667,276,344,607	0.023406	1.023982
10^{20}	2,220,819,602,560,918,840	0.022208	1.022725

Prime counts are exact. Ratios are rounded to six decimal places.

Sources: OEIS A006880 (oeis.org/A006880); primecount.

$\pi(x)$: a step function

Exact graph for $0 \le x \le 100$ on ordinary horizontal and vertical scales.

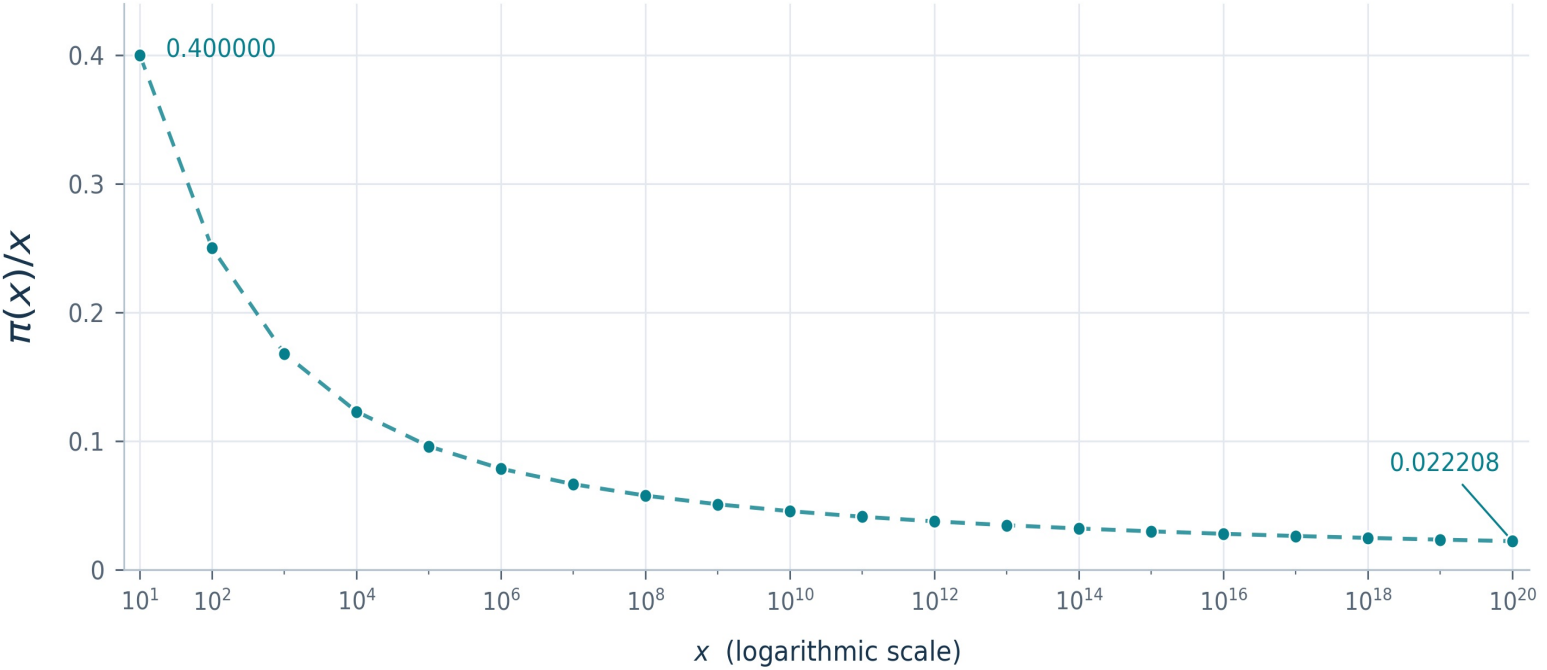


Filled dots give $\pi(p)$ at a prime p; open dots mark the excluded lower endpoints.

Dashed vertical lines indicate the jumps. Each jump has height 1.

The proportion of primes

Exact samples at $x = 10^k$, $k = 1, \ldots, 20$; $\ln x$ is the natural logarithm.



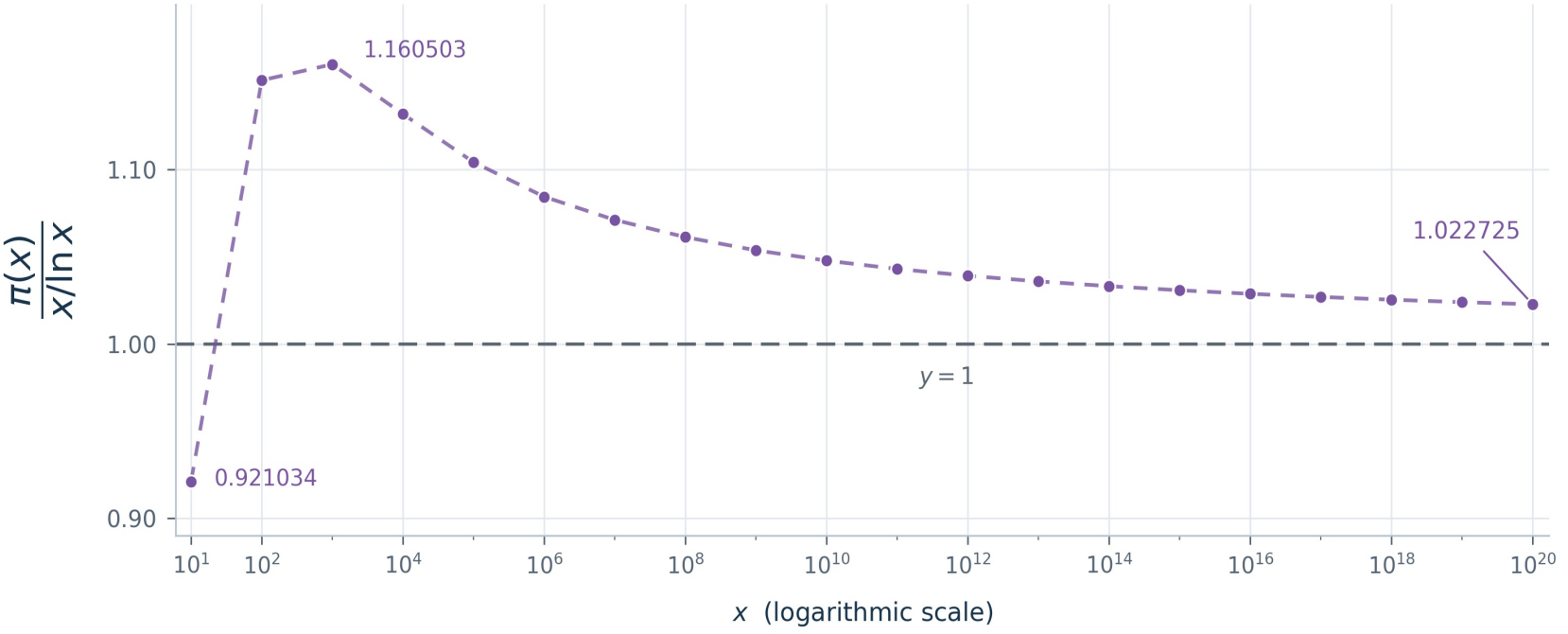
Dots are exact sampled values. Dashed connecting lines are only guides to the eye.

The full function has small jumps at primes; this picture shows its large-scale trend.

Prime counts: OEIS A006880 and the primecount benchmark table.

The normalized prime count

Exact samples at $x = 10^k$, $k = 1, \ldots, 20$; $\ln x$ is the natural logarithm.



Dots are exact sampled values. Dashed connecting lines are only guides to the eye.

The full function has small jumps at primes; this picture shows its large-scale trend.

Prime counts: OEIS A006880 and the primecount benchmark table.

Legendre conjectured

$$\lim_{x\to\infty} \frac{\pi(x)}{\frac{x}{\ln x - 1.08\ldots}} = 1.$$

He meant

$$\pi(x) \approx \frac{x}{\ln x - 1.08\ldots} > \frac{x}{\ln x}.$$

Gauss suggested the logarithmic integral function

$$li(x) = \int_2^x \frac{1}{\ln t}\, dt$$

$$= \frac{x}{\ln x} + \frac{x}{(\ln x)^2} + \text{constant} + 2\int_2^x \frac{1}{(\ln t)^3}\, dt$$

$$> \frac{x}{\ln x}, \quad x \geq 3.$$

Prime-counting approximations: values

$\pi(x) = \#\{p \le x : p \text{ is prime}\}$ $\quad\quad$ $\mathrm{li}(x) = \int_2^x \frac{dt}{\ln t}$

x	$\pi(x)$	$x/\ln x$	$\mathrm{li}(x)$
10^1	4	4.343	5.120
10^2	25	21.715	29.081
10^3	168	144.765	176.564
10^4	1,229	1,085.736	1,245.092
10^5	9,592	8,685.890	9,628.764
10^6	78,498	72,382.414	78,626.504
10^7	664,579	620,420.688	664,917.360
10^8	5,761,455	5,428,681.024	5,762,208.330
10^9	50,847,534	48,254,942.434	50,849,233.912
10^{10}	455,052,511	434,294,481.903	455,055,613.541
10^{11}	4,118,054,813	3,948,131,653.666	4,118,066,399.576
10^{12}	37,607,912,018	36,191,206,825.271	37,607,950,279.760
10^{13}	346,065,536,839	334,072,678,387.117	346,065,645,809.005
10^{14}	3,204,941,750,802	3,102,103,442,166.084	3,204,942,065,690.909
10^{15}	29,844,570,422,669	28,952,965,460,216.789	29,844,571,475,286.536
10^{16}	279,238,341,033,925	271,434,051,189,532.392	279,238,344,248,555.748
10^{17}	2,623,557,157,654,233	2,554,673,422,960,304.869	2,623,557,165,610,820.733
10^{18}	24,739,954,287,740,860	24,127,471,216,847,323.758	24,739,954,309,690,413.977
10^{19}	234,057,667,276,344,607	228,576,043,106,974,646.132	234,057,667,376,222,381.178
10^{20}	2,220,819,602,560,918,840	2,171,472,409,516,259,138.256	2,220,819,602,783,663,482.503

Prime counts are exact. Approximation values are rounded to three decimal places.

Sources: OEIS A006880 (oeis.org/A006880); primecount. Calculations use high-precision arithmetic.

Prime-counting approximations: errors

Relative error of $A(x)$: $\dfrac{|A(x)-\pi(x)|}{\pi(x)}\times 100\%$

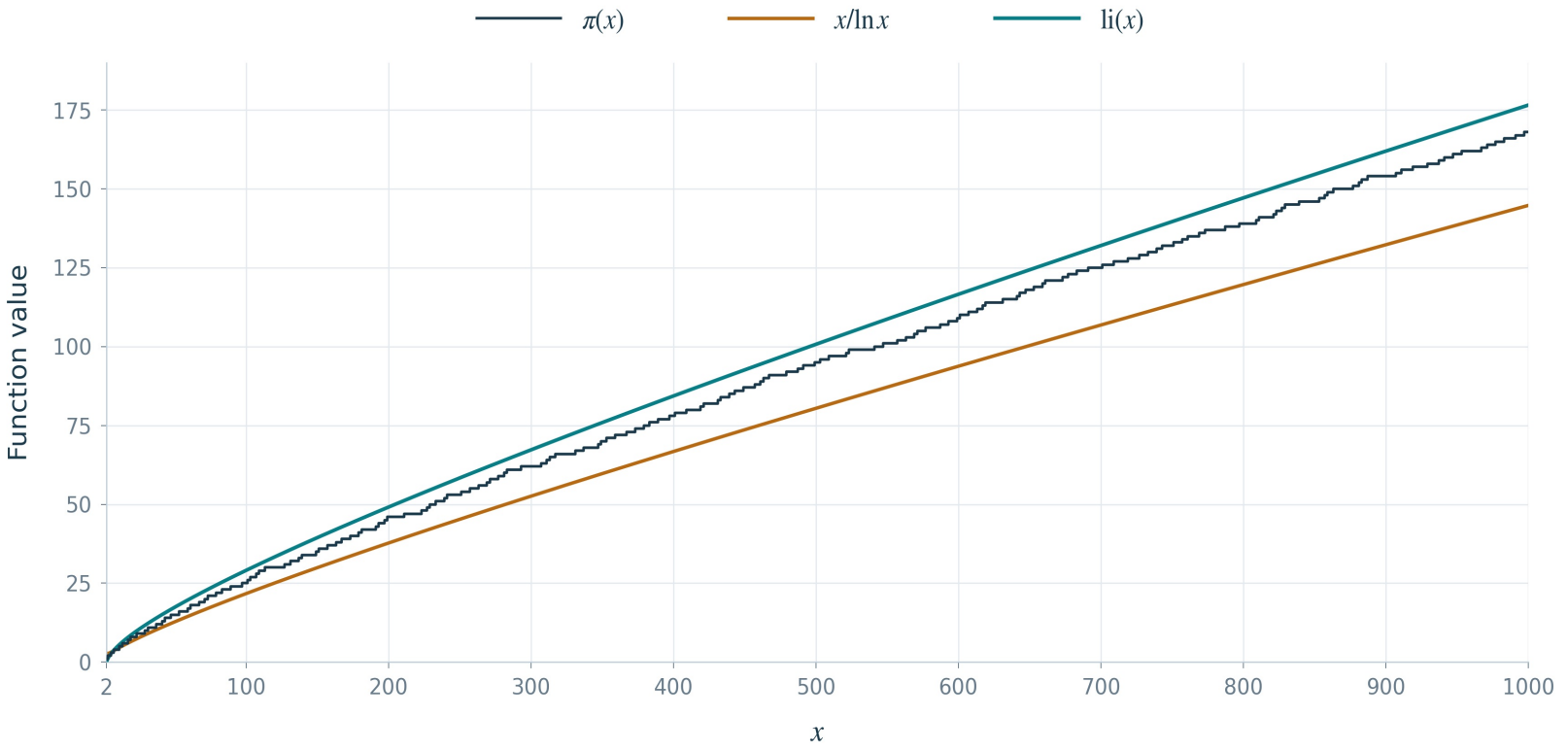
x	$\pi(x)-x/\ln x$	$\pi(x)-\mathrm{li}(x)$	Relative error (%) $x/\ln x$	Relative error (%) $\mathrm{li}(x)$
10^1	−0.343	−1.120	8.5736	28.011
10^2	3.285	−4.081	13.141	16.324
10^3	23.235	−8.564	13.830	5.0979
10^4	143.264	−16.092	11.657	1.3094
10^5	906.110	−36.764	9.4465	0.38328
10^6	6,115.586	−128.504	7.7908	0.16370
10^7	44,158.312	−338.360	6.6446	0.050913
10^8	332,773.976	−753.330	5.7759	0.013075
10^9	2,592,591.566	−1,699.912	5.0988	0.0033432
10^{10}	20,758,029.097	−3,102.541	4.5617	6.8180×10^{-4}
10^{11}	169,923,159.334	−11,586.576	4.1263	2.8136×10^{-4}
10^{12}	1,416,705,192.729	−38,261.760	3.7670	1.0174×10^{-4}
10^{13}	11,992,858,451.883	−108,970.005	3.4655	3.1488×10^{-5}
10^{14}	102,838,308,635.916	−314,888.909	3.2087	9.8251×10^{-6}
10^{15}	891,604,962,452.211	−1,052,617.536	2.9875	3.5270×10^{-6}
10^{16}	7,804,289,844,392.608	−3,214,630.748	2.7948	1.1512×10^{-6}
10^{17}	68,883,734,693,928.131	−7,956,587.733	2.6256	3.0327×10^{-7}
10^{18}	612,483,070,893,536.242	−21,949,553.977	2.4757	8.8721×10^{-8}
10^{19}	5,481,624,169,369,960.868	−99,877,774.178	2.3420	4.2672×10^{-8}
10^{20}	49,347,193,044,659,701.744	−222,744,642.503	2.2220	1.0030×10^{-8}

Differences are signed and rounded to three decimal places. Relative errors use absolute values and five significant figures.

Sources: OEIS A006880 (oeis.org/A006880); primecount. Calculations use high-precision arithmetic.

Prime-counting approximations on $2 \le x \le 1000$

$\mathrm{li}(x) = \int_2^x \frac{dt}{\ln t}$ | Linear axes



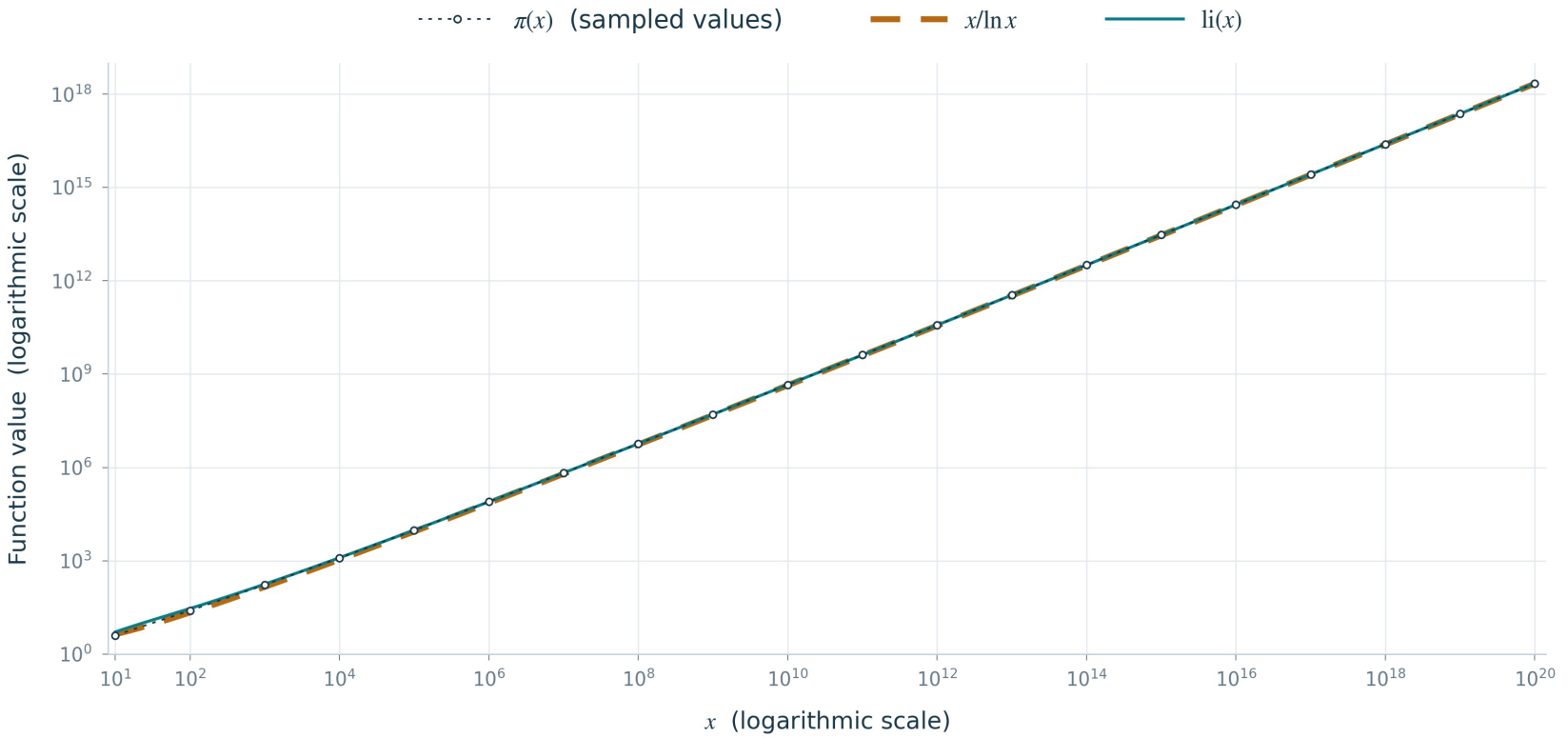
$\pi(x)$ is shown as its full step function, with a jump at every prime.

The curves $x/\ln x$ and $\mathrm{li}(x)$ are evaluated continuously across the interval.

Prime counts computed exactly by the sieve of Eratosthenes.

Prime-counting approximations up to 10^{20}

$\mathrm{li}(x) = \int_2^x \frac{dt}{\ln t}$ | $10 \le x \le 10^{20}$



Dots give exact prime counts at $x = 10^k$, $k = 1, \ldots, 20$; dashed connecting segments are guides to the eye.

Prime counts: OEIS A006880 and primecount. The two approximations are plotted as smooth curves.

Conjecture (Gauss): $\pi(x) \approx li(x)$ better than $x/\ln x$.

Of course,

$$\lim_{x\to\infty} \frac{\pi(x)}{li(x)} = 1 \iff \lim_{x\to\infty} \frac{\pi(x)}{x/\ln x} = 1.$$

Ex. Show this by proving

$$\lim_{x\to\infty} \frac{li(x)}{x/\ln x} = 1.$$

Prime Number Theorem:

$$\lim_{x\to\infty} \frac{\pi(x)}{x/\ln x} = 1.$$

We will prove that there are constants $C_1, C_2 > 0$

s.t.

$$C_1 \frac{x}{\ln x} < \pi(x) < C_2 \frac{x}{\ln x}, \quad x \geq 2.$$

Lower bound: Give $N \geq 2$, is there a naive choice

of number that is divisible by every $p \leq N$?

Maybe $N!$, since every $p \leq N$ divides $N!$.

Moreover, $p > N \Rightarrow p \nmid N!$

Shortcomings. 1. $N!$ is too large!

2. May have $p^k \mid N!$ with k large.

Candidate: $\binom{2N}{N} = \frac{(2N)!}{N! \cdot N!} \in \mathbb{Z}$.

1. $p \mid \binom{2N}{N} \Rightarrow p \leq 2N$.

2. Size of $\binom{2N}{N}$ is more manageable.

3. There is a chance that large prime powers (excluding primes) do not divide $\binom{2N}{N}$.

Will use this idea next time.